Neural communication

Neurons communicate through electrical and chemical synapses

Computational Neuroscience

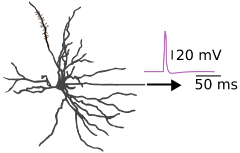
University of Bristol

M Rule

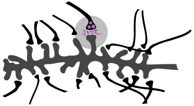
Learning outcomes:

- What is a synapse?
- Key terms
- Steps of synaptic transmission
- Describe what synapses are and how they work
- Explain the steps of electrochemical transmission in synapse

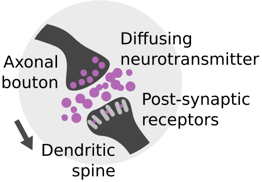
Information flow in single neurons



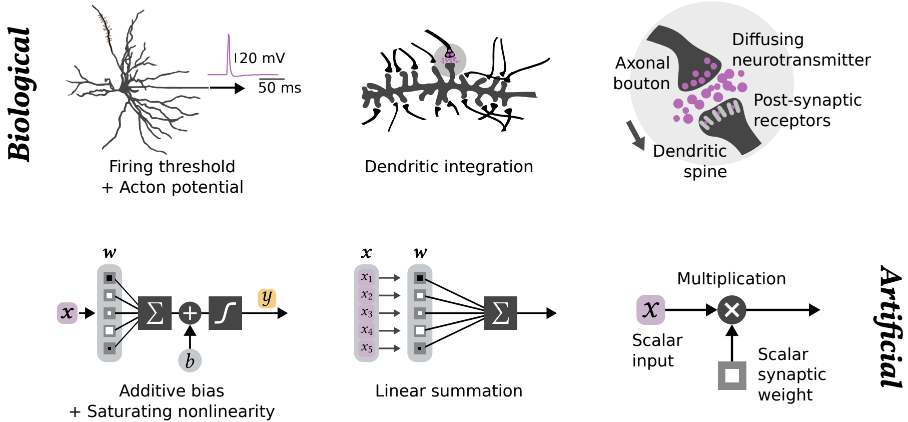
Firing threshold
+ Acton potential

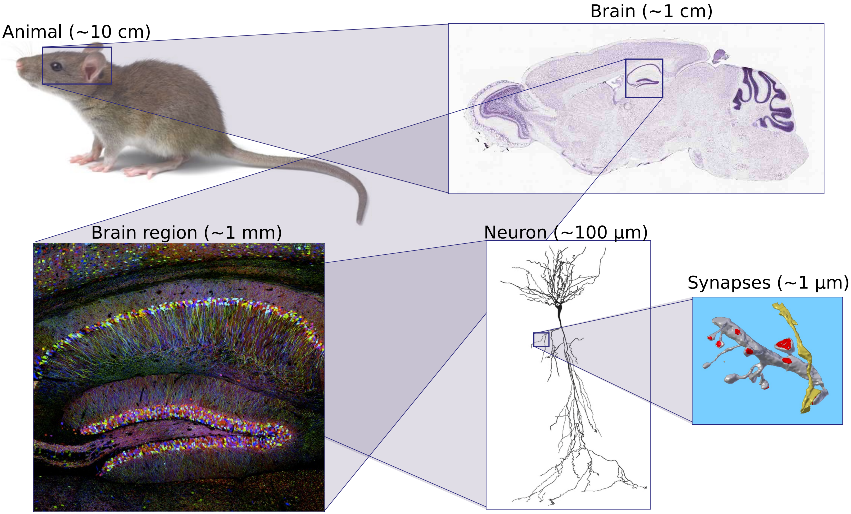


Dendritic integration



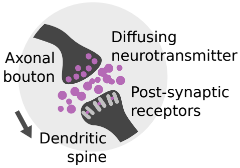
Information flow in single neurons



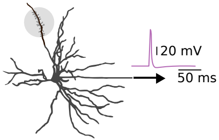


Modified from B. Mizusaki (2022) lecture slides

Neurons communicate through electrical and chemical synapses



Dendritic integration



Firing threshold
+ Acton potential

Electrical synapses (“gap junctions”) connect the interiors of two neurons.

ask me about Camillo Golgi and the syncytium

Chemical synapses

- $\sim 1\,\mu\mathrm{m}$ in size;
- Δ synaptic strength $\rightarrow$ learning, memory
- **Pre**synaptic spike $\rightarrow$ **Post**synaptic potential (PSP)
- Sometimes unreliable
- May have complicated, time-dependent, nonlinear dynamics (some computation happens here)

Excitatory

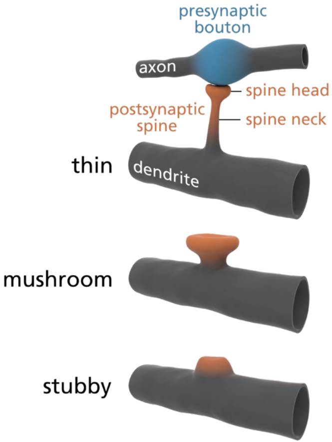
- Make targets more likely to spike
- **Glutamate** (others)

Inhibitory

- Make targets less likely to spike
- **GABA** (others)

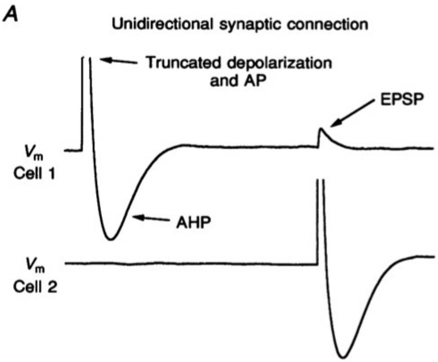
(and repeat)

(many exceptions to every statement here)



Splettstoesser (2014) via Wikimedia

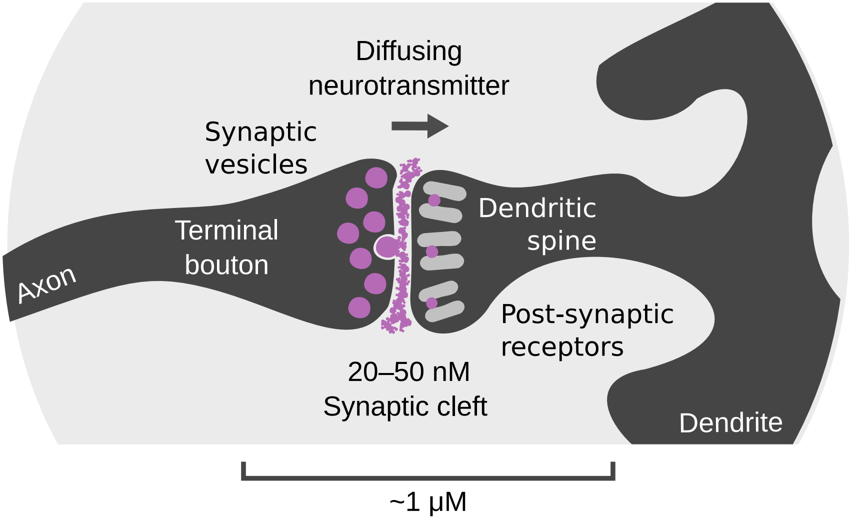
Chemical synapses are ≈ unidirectional



Markram et al, J Physiol (1997)

synapse → dendrite → soma → axon → synapse

Chemical synapses are ≈ unidirectional



Add calcium channels, reuptake pumps, label intra/extracellular spaces draw a synapse, animate the steps of synaptic transmission

What is a synapse?

A connection between neurons that converts the action potential from a presynaptic neuron's axon (output) into a post-synaptic potential that acts as input for a another, postsynaptic neuron.

What kind of connection is a chemical synapse?

An electrochemical connections between neuronal cells.

- They can be excitatory, inhibitory (modulatory, other...)
- They are nonlinear, so can perform computations.
- They are plastic, so can store information (learning, memory)
- Neurotransmitter release is unreliable, so a synapse can be considered a probabilistic form of communication.

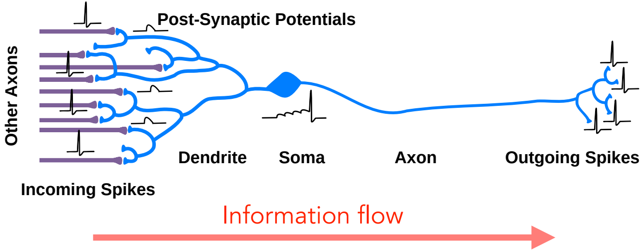
Steps in synaptic transmission

a. *Presynaptic*
   1. Action potential arrives
   2. Voltage gated calcium channels open
   3. Ca^{2+} enters the presynaptic cell
   4. Ca^{2+} triggers vesicle fusion

b. ((Neurotransmitter diffuses across synapse))

c. *Postsynaptic*
   1. Neurotransmitter binds to receptors on postsynaptic membrane
   2. Ionotropic receptors open, allowing ionic currents
      Metabotropic receptors trigger signalling cascades that modify ionic conductances
   3. Ionic currents change dendritic voltage

Metabotropic receptors: G-Protein Coupled Receptors (GPCRs), Receptor-Tyrosine Kinases (RTKs), Guanylate Cyclase Receptors, ...



1. A pulse of neurotransmitters is released into synaptic cleft and binds to postsynaptic receptors
2. Receptors open, causing ionic currents that change the postsynaptic voltage ~ 1 mV for ~ 10 ms

   ↑ **Post-Synaptic Potential** **PSP**
3. Voltages sum at the soma; if $V_{\text{soma}} > V_{\text{threshold}}$, the cell spikes.
4. This spike travels out the axon triggering neurotransmitter release from axonal terminals

   (repeat)

draw a synapse

ligand/voltage-gated, EPSP/IPSP

end

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... Quantum Chemistry, Molecular dynamics

Physiological, Quantitative

Biological Realism, Data needed to identify parameters

Molecules	Gillespie, Master Equation
Concentrations	Mass-Action Kinetics
Conductance Models	Hodgkin–Huxley
Spiking Models	Leaky Integrate and Fire
Rate Neurons	Neural Mass/Field Models
Poisson Neurons	Generalized Linear Models
Binary Neurons	McCulloch–Pitts, Hopfield, Perceptron

Computational Efficiency, Mathematical Tractability

Phenomenological, Qualitative

Cognitive Neuroscience, Psychology ...